Bayesian vs Frequentist Decision Frameworks
Connections via LDA, Neyman–Pearson, and Logistic Regression

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Goal: Understand how different statistical philosophies lead to similar decision rules.

Three frameworks:

1. **Bayesian:** Posterior probabilities
2. **Frequentist (Neyman–Pearson):** Likelihood ratios
3. **Statistical Learning:** Logistic regression

All lead to the same decision boundary under certain assumptions

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Philosophical Foundations.

Frequentist

- Parameters are fixed
- Probability = long-run frequency
- Focus: hypothesis testing & error control

Bayesian

- Parameters are random
- Probability = degree of belief
- Focus: posterior inference

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Binary Classification Setup

Let:

$$Y \in \{0, 1\}, \qquad X = x$$

Goal:

Classify x *into class* 0 *or* 1

Bayesian Decision Rule

Using Bayes rule:

$$P(Y = 1 \mid x) = \frac{\pi_1 f_1(x)}{\pi_0 f_0(x) + \pi_1 f_1(x)}$$

Decision rule:

$$\text{Choose } 1 \ \textit{ if } P(Y = 1 \mid x) > 0.5$$

Equivalent:

$$\frac{\pi_1 f_1(x)}{\pi_0 f_0(x)} > 1$$

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Neyman–Pearson Lemma

Test:

$$H_0: f_0(x) \; vs \; H_1: f_1(x)$$

Optimal test:

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)} > c$$

Most powerful test for fixed Type I error

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Bayesian vs Neyman–Pearson

Bayesian rule:

$$\frac{\pi_1 f_1(x)}{\pi_0 f_0(x)} > 1$$

Rewrite:

$$\frac{f_1(x)}{f_0(x)} > \frac{\pi_0}{\pi_1}$$

Same form as Neyman–Pearson!

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Key Connection

Bayesian *classification* = *Likelihood ratio test*

Difference:
Bayesian: threshold depends on priors
Frequentist: threshold depends on Type I error

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Linear Discriminant Analysis (LDA)

Assume:

$$X \mid Y = k \sim N(\mu_k, \sigma^2)$$

Discriminant function:

$$\delta_k(x) = \frac{\mu_k}{\sigma^2} x - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k$$

Decision:

Choose *class* $k = \arg\max \delta_k(x)$

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LDA = Likelihood Ratio

Compute:

$$\log \frac{f_1(x)}{f_0(x)}$$

Result:

$$= \frac{\mu_1 - \mu_0}{\sigma^2} x + constant$$

Linear in x

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LDA Decision Rule

$$\delta_1(x) - \delta_0(x) = \log \frac{\pi_1 f_1(x)}{\pi_0 f_0(x)}$$

Decision:

Choose 1 *if* $\delta_1(x) > \delta_0(x)$

Equivalent to likelihood ratio test

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Posterior from LDA

$$P(Y = 1 \mid x) = \frac{e^{\delta_1(x)}}{e^{\delta_0(x)} + e^{\delta_1(x)}}$$

Rewrite:

$$P(Y = 1 \mid x) = \frac{1}{1 + e^{-\left(\delta_1(x) - \delta_0(x)\right)}}$$

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Logistic Form Appears

Let:

$$\delta_1(x) - \delta_0(x) = \beta_0 + \beta_1 x$$

Then:

$$\boxed{P(Y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}}$$

This is logistic regression

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Logistic Regression Connection

Logistic model:

$$\log \frac{P(Y = 1 \mid x)}{P(Y = 0 \mid x)} = \beta_0 + \beta_1 x$$

Same as:

$$\log \frac{\pi_1 f_1(x)}{\pi_0 f_0(x)}$$

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Unification

All three approaches yield:

Decision *rule based on* $\log \frac{f_1(x)}{f_0(x)}$

Framework	**Interpretation**
Bayesian	Posterior probability (probability of class)
Neyman–Pearson	Likelihood ratio (optimal test)
Logistic Regression	Log-odds model (predictive model)

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Summary

Bayesian:

$$P(Y = 1 \mid x)$$

Neyman–Pearson:

$$\frac{f_1(x)}{f_0(x)}$$

Logistic regression:

$$\log \frac{P(Y = 1 \mid x)}{P(Y = 0 \mid x)}$$

All lead to the same linear decision boundary

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Conclusions

Statistics *is unified by decision theory — philosophy only changes interpretation*.

LDA, logistic regression, and hypothesis testing are all solving the same problem — deciding between two distributions — just from different philosophical angles.

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References

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Final Remarks and Acknowledgement

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Miscellaneous R Code

Posterior Probability from LDA Scores

```
delta0 <- -0.5
delta1 <- 1.2
p1 <- exp(delta1) / (exp(delta0) + exp(delta1))
```

Log-Odds and Logistic Transformation

```
p <- 0.7
log_odds <- log(p / (1 - p))
logistic <- function(z) {1/(1+exp(-z))}
```

Likelihood Ratio (Neyman–Pearson)

```
f0 <- dnorm(1,0,1)
f1 <- dnorm(1,2,1)
LR <- f1/f0
```

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Miscellaneous R Code

Univariate LDA Discriminant Functions

```
x<-1; mu0<-0; mu1<-2; s2<-1
d0<-x*mu0/s2 - mu0^2/(2*s2) + log(0.5)
d1<-x*mu1/s2 - mu1^2/(2*s2) + log(0.5)
```

Simulating Two Gaussian Classes

```
set.seed(1)
x0<-rnorm(100,0,1)
x1<-rnorm(100,2,1)
y<-factor(c(rep(0,100),rep(1,100)))
x<-c(x0,x1)
```

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Miscellaneous R Code

Logistic Regression Model

```
dat<-data.frame(x,y)
fit<-glm(y~x,dat,family=binomial)
summary(fit)
```

Predicted Probabilities

```
prob<-predict(fit,type="response")
head(prob)
```

LDA Model and Posterior

```
library(MASS)
fit_lda<-lda(y~x,dat)
pred<-predict(fit_lda)
head(pred$posterior)
```

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Miscellaneous R Code

Decision Boundaries

```
b0<-coef(fit)[1]
b1<-coef(fit)[2]
x_logit<- -b0/b1
x_lda<- (mean(x0)+mean(x1))/2
```

Logistic Curve Visualization

```
xg<-seq(-4,6,len=200)
pg<-predict(fit,data.frame(x=xg),type="response")
plot(x,y); lines(xg,pg)
```

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